

Financial Statements?

- Components of Financial statement
- Accounting income vs. cash flow
- Ratio Analysis
- Effects of improving ratios
- Limitations of ratio analysis
- Qualitative factors

The Annual Report

- **Balance sheet** – provides a snapshot of a firm's financial position at one point in time.
- **Income statement** – summarizes a firm's revenues and expenses over a given period of time.
- **Statement of retained earnings** – shows how much of the firm's earnings were retained, rather than paid out as dividends. (*how does it effect to dividend policy?*)
- **Statement of cash flows** – reports the impact of a firm's activities on cash flows over a given period of time. (Cash inflows and cash out flows)

Balance Sheet: Assets

	<u>2017</u>	<u>2018</u>
Cash	7,282	57,600
A/R	632,160	351,200
Inventories	<u>1,287,360</u>	<u>715,200</u>
Total CA	1,926,802	1,124,000
Gross FA	1,202,950	491,000
Less: Dep.	<u>263,160</u>	<u>146,200</u>
Net FA	<u>939,790</u>	<u>344,800</u>
Total Assets	<u><u>2,866,592</u></u>	<u><u>1,468,800</u></u>

Balance sheet: Liabilities and Equity

	2017	2018
Accts payable	<u>524,160</u>	<u>145,600</u>
Notes payable	636,808	200,000
Accruals	<u>489,600</u>	<u>136,000</u>
Total CL	1,650,568	481,600
Long-term debt	723,432	323,432
Common stock	460,000	460,000
Retained earnings	<u>32,592</u>	<u>203,768</u>
Total Equity	<u>492,592</u>	<u>663,768</u>
Total L & E	<u><u>2,866,592</u></u>	<u><u>1,468,800</u></u>

Income statement

	2017	2018
Sales	6,034,000	3,432,000
COGS	5,528,000	2,864,000
Other expenses	<u>519,988</u>	<u>358,672</u>
EBITDA	(13,988)	209,328
Depr. & Amort.	<u>116,960</u>	<u>18,900</u>
EBIT	(130,948)	190,428
Interest Exp.	<u>136,012</u>	<u>43,828</u>
EBT	(266,960)	146,600
Taxes	<u>(106,784)</u>	<u>58,640</u>
Net income	<u><u>(160,176)</u></u>	<u><u>87,960</u></u>

Other data

	<u>2017</u>	<u>2018</u>
No. of shares	100,000	100,000
EPS	-\$1.602	\$0.88
DPS	\$0.11	\$0.22
Stock price	\$2.25	\$8.50
Lease pmts	\$40,000	\$40,000

Statement of Retained Earnings (2018)

Balance of retained earnings, 12/31/17	\$203,768
Add: Net income, 2018	(160,176)
Less: Dividends paid	<u>(11,000)</u>
Balance of retained earnings, 12/31/18	<u><u>\$32,592</u></u>

Statement of Cash Flows (2018)

OPERATING ACTIVITIES

Net income	(160,176)
Add (Sources of cash):	
Depreciation	116,960
Increase in A/P	378,560
Increase in accruals	353,600
Subtract (Uses of cash):	
Increase in A/R	(280,960)
Increase in inventories	<u>(572,160)</u>
Net cash provided by ops.	(164,176)

Statement of Cash Flows (2018)

L-T INVESTING ACTIVITIES

Investment in fixed assets	(711,950)
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FINANCING ACTIVITIES

Increase in notes payable	436,808
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Increase in long-term debt	400,000
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Payment of cash dividend	<u>(11,000)</u>
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Net cash from financing	825,808
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NET CHANGE IN CASH	(50,318)
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Plus: Cash at beginning of year	<u>57,600</u>
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Cash at end of year	<u><u>7,282</u></u>
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What can you conclude about the financial condition from its statement of CFs?

- Net cash from operations = -\$164,176, mainly because of negative NI.
- The firm borrowed \$825,808 to meet its cash requirements.
- Even after borrowing, the cash account fell by \$50,318.

Did the expansion create additional net operating after taxes (NOPAT)?

Thus, $EBIT_{18} = -\$130,948$, $Tax = 40\%$, $NOPAT_{18} = ??$

$$NOPAT = EBIT (1 - Tax\ rate)$$

$$\begin{aligned} NOPAT_{18} &= -\$130,948(1 - 0.4) \\ &= -\$130,948(0.6) \\ &= -\$78,569 \end{aligned}$$

$$NOPAT_{17} = \$114,257$$

What effect did the expansion have on net operating working capital?

CA = Cash + A/C receivable + Inventories

CL = A/C payable + Notes payable + Accruals

NOWC = Current assets - Non-interest bearing CL

$$\begin{aligned}\text{NOWC}_{18} &= (\$7,282 + \$632,160 + \$1,287,360) \\ &\quad - (\$524,160 + \$489,600) \\ &= \$913,042\end{aligned}$$

$$\text{NOWC}_{17} = \$842,400$$

What effect did the expansion have on operating capital?

Operating capital = NOWC + Net Fixed Assets

$$\begin{aligned}\text{Operating Capital}_{18} &= \$913,042 + \$939,790 \\ &= \$1,852,832\end{aligned}$$

$$\text{Operating Capital}_{17} = \$1,187,200$$

What is your assessment of the expansion's effect on operations?

	2018	2017
Sales	<u>\$6,034,000</u>	<u>\$3,432,000</u>
NOPAT	-\$78,569	\$114,257
NOWC	\$913,042	\$842,400
Operating capital	\$1,852,832	\$1,187,200
Net Income	-\$160,176	\$87,960

What effect did the expansion have on net cash flow and operating cash flow?

$$\begin{aligned} \text{NCF}_{18} &= \text{NI} + \text{Dep} = (\$160,176) + \$116,960 \\ &= -\$43,216 \end{aligned}$$

$$\text{NCF}_{17} = \$87,960 + \$18,900 = \$106,860$$

$$\begin{aligned} \text{OCF}_{18} &= \text{NOPAT} + \text{Dep} \\ &= (\$78,569) + \$116,960 \\ &= \$38,391 \end{aligned}$$

$$\begin{aligned} \text{OCF}_{17} &= \$114,257 + \$18,900 \\ &= \$133,157 \end{aligned}$$

What was the free cash flow (FCF) for 2018?

$FCF = OCF - \text{Gross capital investment}$

- OR -

$$\begin{aligned} FCF_{18} &= \text{NOPAT} - \text{Net capital investment} \\ &= -\$78,569 - (\$1,852,832 - \$1,187,200) \\ &= -\$744,201 \end{aligned}$$

Is negative cash flow always a bad sign?

Financial Ratio

Why are financial ratios useful?

- Ratios standardize numbers and facilitate comparisons. (Financial statement analysis?)
- Ratios are used to highlight weaknesses and strengths.

What are the five major categories of ratios, and what questions do they answer?

- **Liquidity:** Can we make required payments?
- **Asset management:** right amount of assets vs. sales?
- **Debt management:** Right mix of debt and equity?
- **Profitability:** Do sales prices exceed unit costs, and are sales high enough as reflected in PM, ROE, and ROA?
- **Market value:** Do investors like what they see as reflected in P/E and M/B ratios?

Comments on current ratio (example)

	2018	2017	2016	Ind.
Current ratio	2.34x	1.20x	2.30x	2.70x

- Expected to improve but still below the industry average.
- Liquidity position is weak.

$$\text{CR} = \text{CA}/\text{CL}, \text{QR} = (\text{CA} - \text{Inventory})/\text{CL}$$

What is the inventory turnover vs. the industry average?

Given, Sales= \$7,036 , Inventories= \$1,716

$$\begin{aligned}\text{Inv. turnover} &= \text{Sales} / \text{Inventories} \\ &= \$7,036 / \$1,716 \\ &= 4.10x\end{aligned}$$

	2018	2017	2016	Ind.
Inventory Turnover	4.1x	4.70x	4.8x	6.1x

What is DSO?

DSO is the average number of days after making a sale before receiving cash.

Given, Receivables= \$878 , Average sales per day=
\$7,036/365

$$\begin{aligned}\text{DSO} &= \text{Receivables} / \text{Average sales per day} \\ &= \text{Receivables} / \text{Sales}/365 \\ &= \$878 / (\$7,036/365) \\ &= 45.6\end{aligned}$$

Appraisal of DSO

	2018	2017	2016	Ind.
DSO	45.6	38.2	37.4	32.0

- The firm collects on sales too slowly, and is getting worse.
- The firm has a poor credit policy.

Fixed asset and total asset turnover ratios vs. the industry average

Given, Sales= \$7,036 , Net fixed assets= \$817, Total
assets= \$3,497

$$\begin{aligned}\text{FA turnover} &= \text{Sales} / \text{Net fixed assets} \\ &= \$7,036 / \$817 = 8.61x\end{aligned}$$

$$\begin{aligned}\text{TA turnover} &= \text{Sales} / \text{Total assets} \\ &= \$7,036 / \$3,497 = 2.01x\end{aligned}$$

Evaluating the FA turnover and TA turnover ratios and their meaning

	2018	2017	2016	Ind.
FA TO	8.6x	6.4x	10.0x	7.0x
TA TO	2.0x	2.1x	2.3x	2.6x

Do you think the company is performing well?

- FA turnover projected to exceed the industry average.
- TA turnover below the industry average.
Caused by excessive current assets (A/R and Inv).

Calculate the debt ratio, TIE, and EBITDA coverage ratios.

Given, Short-term debt= \$1,145, long-term debt= \$400, Total assets= \$3,497

$$\begin{aligned}\text{Debt ratio} &= \text{Total debt} / \text{Total assets} \\ &= (\$1,145 + \$400) / \$3,497 = 44.2\%\end{aligned}$$

$$\begin{aligned}\text{TIE} &= \text{EBIT} / \text{Interest expense} \\ &= \$492.6 / \$70 = 7.0x\end{aligned}$$

$$\begin{aligned}\text{EBITDA coverage} &= \frac{(\text{EBITDA} + \text{Lease pmts})}{\text{Int exp} + \text{Lease pmts} + \text{Principal pmts}} \\ &= \frac{\$609.6 + \$40}{\$70 + \$40 + \$0} \\ &= 5.9x\end{aligned}$$

How do the debt management ratios compare with industry averages?

	2018	2017	2016	Ind.
D/A	44.2%	82.8%	54.8%	50.0%
TIE	7.0x	-1.0x	4.3x	6.2x
EBITDA coverage	5.9x	0.1x	3.0x	8.0x

- D/A and TIE are better than the industry average, but EBITDA coverage still trails the industry.

Profitability ratios:

Profit margin and Basic earning power

$$\begin{aligned}\text{Profit margin} &= \text{Net income} / \text{Sales} \\ &= \$253.6 / \$7,036 = 3.6\%\end{aligned}$$

$$\begin{aligned}\text{BEP} &= \text{EBIT} / \text{Total assets} \\ &= \$492.6 / \$3,497 = 14.1\%\end{aligned}$$

Appraising profitability with the profit margin and basic earning power

	2018	2017	2016	Ind.
PM	3.6%	-2.7%	2.6%	3.5%
BEP	14.1%	-4.6%	13.0%	19.1%

- Profit margin was very bad in 2017, but is projected to exceed the industry average in 2018. Looking good.
- BEP removes the effects of taxes and financial leverage, and is useful for comparison.
- BEP projected to improve, yet still below the industry average. There is definitely room for improvement.

Profitability ratios:

Return on assets and Return on equity

$$\begin{aligned}\text{ROA} &= \text{Net income} / \text{Total assets} \\ &= \$253.6 / \$3,497 = 7.3\%\end{aligned}$$

$$\begin{aligned}\text{ROE} &= \text{Net income} / \text{Total common equity} \\ &= \$253.6 / \$1,952 = 13.0\%\end{aligned}$$

Appraising profitability with the return on assets and return on equity

	2018	2017	2016	Ind.
ROA	7.3%	-5.6%	6.0%	9.1%
ROE	13.0%	-32.5%	13.3%	18.2%

- Both ratios rebounded from the previous year, but are still below the industry average. More improvement is needed.
- Wide variations in ROE illustrate the effect that leverage can have on profitability.

Calculate the Price/Earnings, Price/Cash flow, and Market/Book ratios.

$$\begin{aligned}\text{P/E} &= \text{Price} / \text{Earnings per share} \\ &= \$12.17 / \$1.014 = 12.0x\end{aligned}$$

$$\begin{aligned}\text{P/CF} &= \text{Price} / \text{Cash flow per share} \\ &= \$12.17 / [(\$253.6 + \$117.0) \div 250] \\ &= 8.21x\end{aligned}$$

Calculate the Price/Earnings, Price/Cash flow, and Market/Book ratios.

$$\begin{aligned} \text{M/B} &= \text{Mkt price per share} / \text{Book value per share} \\ &= \$12.17 / (\$1,952 / 250) = 1.56x \end{aligned}$$

	2018	2017	2016	Ind.
P/E	12.0x	-1.4x	9.7x	14.2x
P/CF	8.21x	-5.2x	8.0x	11.0x
M/B	1.56x	0.5x	1.3x	2.4x

Potential uses of freed up cash

- Repurchase stock
- Expand business
- Reduce debt
- All these actions would likely improve the stock price.

Potential problems and limitations of financial ratio analysis

- Comparison with industry averages is difficult for a conglomerate firm that operates in many different divisions.
- “Average” performance is not necessarily good, perhaps the firm should aim higher.
- Seasonal factors can distort ratios.
- “Window dressing” techniques can make statements and ratios look better.

(Discussion on Term paper and Financial Statement)